

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE			
Supplementary Examination –Summer 2024			
Course: B. Tech. Branch : Electrical Engineering and Allied Semester : V			
Subject Code & Name: BTEEC502 Microprocessor and Microcontroller			
Max Marks: 60	Date: 03/07/2024	Duration: 3 Hr.	
Instructions to the Students:			
1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Attempt any two (2) of the following.		12
A)	"The ALE signal is an important factor in preserving the contents of A0-A7 lines." Justify your answer with suitable statements.		6
B)	Define addressing modes and explain each mode with suitable examples.		6
C)	Discuss the bit pattern of 8085 flag register and specify the contents of flags after execution of the following instruction XRA D. Assume [A] = 77h and [D] = 06h.		6
Q.2	Attempt any two (2) of the following.		12
A)	Define interrupt. Explain the use of TRAP and HOLD signals of 8085 and contrast between them.		6
B)	Write a program to find the sum of all the array elements. The length of the array is stored at memory location BFFFh followed by the data and result.		6
C)	Write a program to count the number of 1's in an 8-bit number stored at memory location C000h. Store the count of 1's in C001h and that of 0's in C002h.		6
Q. 3	Attempt any two (2) of the following.		12
A)	Expand the memory capacity up to 16kB using 8kB memory chip whose starting address is 0000h. Specify the address map and draw the memory organization.		6
B)	Draw and explain the timing diagram for Opcode fetch machine cycle		6
C)	Distinguish between memory mapped I/O and I/O mapped I/O schemes.		6

Q.4	Attempt any two (2) of the following.		12
A)	Explain the ON-chip memory organization of an 8051 microcontroller.		6
B)	Compare microprocessors and microcontrollers with suitable points.		6
C)	Write an assembly language program to find out the sum of two 8-bit numbers stored at memory locations 2000h and 2001h of external RAM. Store the sum in the consecutive next memory locations, assuming the sum may be more than 8-bits.		6
Q. 5	Attempt any two (2) of the following.		12
A)	Define interrupts and describe the various interrupts available in 8051 microcontrollers. Explain the bit format for IE and IP SFRs.		6
B)	Write an assembly language program for generating a square wave of 2kHz frequency on pin P1.5 using mode 1 for timer 1.		6
C)	Explain the bit format for the SCON, TMOD and TCON SFRs.		6
	*** End ***		